InUnity Private Limited
Sahyadri Campus,
Adyar, Mangaluru, Karnataka - 575007

Monthly Progress Report
FEB 23rd - June 13th
2025 - 2026

inunity | innovation for community

Course: Innovation and Design Thinking Course

YENEPOYA
INSTITUTE OF TECHNOLOGY

Report created by:

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InUnity Private Limited @YIT

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Innovation and Design Thinking Course

The Innovation and Design Thinking Program is designed to equip students with the mindset, tools, and hands-on experience to identify, analyze, and solve real-world community challenges. The course blends design thinking, problem-solving frameworks, and prototyping tools to foster creativity, empathy, and entrepreneurial thinking.

Course Objective:

To equip students with the mindset, tools, and methodologies of innovation and design thinking to identify real-world challenges, generate creative solutions, and develop user-centered, impactful prototypes.

Course Outcomes:

Participants of this course will

CO 1	Learn essential prototyping tools and techniques for developing innovative solutions
CO 2	Visualize and draft potential solutions to identified social challenges.
CO 3	Develop and refine prototypes using rapid iteration and testing approaches.

Session Overview:

Course Title	**Innovation and Design Thinking Course**
Year and semester to which this report applies	Academic Year 2025-26, 2nd Semester
Name of Institution SPOC	**Mr. G Jaganesh Chengappa**
Faculty	*Lavitra Sahu - Assistant Program Mentor* *Abhishek Rayareddi - Technical Trainer*
Total no of students enrolled	288
Total no of batches	6

Session Engagement

Current Time Table

Day	9:00-9:50	9:50-10:40	10:40-10:55	10:55-11:45	11:45-12:35	1:45-2:40	2:40-3:35	3:35-4:30
Mon							A Section (AIML)	
Tues				E Section (CSE)				
Wed	F Section(CSE+ISE)					C Section (DS)		
Thurs		G Section (ECE) + D Section (ME)					B Section (IoT)	
Fri	Review							
Sat	Review							

S.No	Session Title	Start Date	End Date	Section Completed
1	Session 1 : Introduction to Design Thinking	23/02/2026	27/02/2026	**Completed**
2	Session 2 : Secondary Research & Mindmapping	02/03/2026	06/03/2026	**Completed**
3	Session 3 : Problem Analysis	09/03/2026	13/03/2026	**Completed**
4	Session 4 : Introduction to Basic Electronics	23/03/2026	26/03/2026	**Completed**
5	Session 5 : CIE Evaluation Phase 1 (Trial)	27/03/2026	28/03/2026	**Completed**
6	Session 6 : Introduction to Microcontrollers	31/03/2026	04/04/2026	**Completed**
7	Session 7: Introduction to Sensor Interfacing and Analog Data Processing Using Arduino	06/04/2026	11/04/2026	**Completed**
8	Session 8: Onshape - Cad Designing	12/04/2026	24/04/2026	**Completed**
9	Session 9: Onshape - Ultimaker Cura - Cad Slicing & 3D Printing	27/04/2026	30/04/2026	**Completed**
10	Session 10: Antigravity - Vibe Coding	27/04/2026	08/04/26	**Completed**
11	Session 11: Easy EDA: PCB Designing	18/05/2026	25/05/25	**Completed**

Session Rating :

Session	Response Count	Session Rating	Faculty Rating
1-3	149	4.79/5	4.72/5
4	131	4.81/5	4.71/5

6	200	4.74/5	4.63/5
7	88	4.78/5	4.70/5
8	86	4.79/5	4.80/5
9	96	4.73/5	4.77/5
10	81	4.78/5	4.79/5
11	115	4.70/5	4.70/5

Session Description:

Session 1 - Introduction to Design Thinking

Session Overview:

Design Thinking is a human-centered approach to problem-solving that focuses on understanding the needs of people, challenging assumptions, and redefining problems to identify innovative solutions. It's not just about creating new products or services, but about building solutions that truly make a difference for users. The process is structured into five key stages:

Empathize – Understand the people you're designing for by observing, engaging, and listening.
Define – Clearly frame the problem based on insights gathered.
Ideate – Generate a wide range of creative ideas and potential solutions.
Prototype – Build simple, testable versions of your ideas.
Test – Try out prototypes with real users, gather feedback, and refine.

Session 2 - Secondary Research and Mindmapping

Session Overview:

Secondary research, also known as desk research, involves collecting and analyzing existing information from sources such as books, journals, reports, and online databases instead of gathering new data. It helps in understanding the background of a problem, exploring existing solutions, and identifying gaps for further study.

It is a cost-effective and time-saving method that is often used as the first step in research, especially in design thinking during the empathize and define stages. However, the information may sometimes be outdated or not fully relevant, and its reliability depends on the credibility of the source.

Session 3 - Problem Analysis

Session Overview:

Session 3 focused on deepening students' understanding of the Empathize phase by introducing structured methods to identify root causes of problems. Building on tools like Desk Research, 5W1H, and Mind Mapping, students learned the 5 Whys technique using examples such as the Titanic sinking to explore multiple layers of causation. They also studied the Fishbone (Ishikawa) Diagram to categorize causes and applied it to real-life scenarios like exam failure.

The session then moved into the Define phase, where students learned to convert insights into opportunity statements using the "How Might We" approach. Through practical examples like improving MRI experiences for children and enhancing ship safety, students understood how empathy and problem analysis lead to innovative solutions.

Session 1,2 & 3 Feedback:

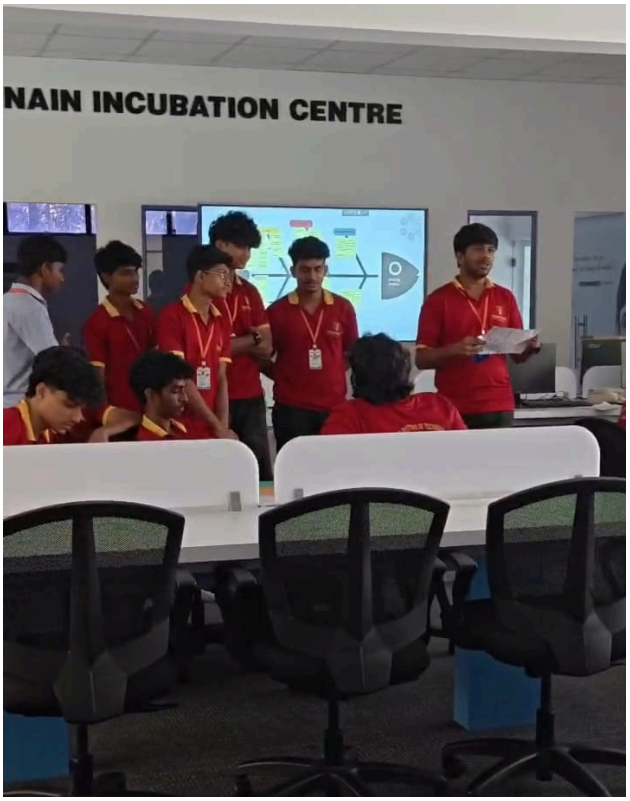
We got around 149 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

- They were good in explaining the activity based projects and good in interacting with all
- Explanation is good
- Got clear picture about what to do in the upcoming future
- Very friendly and helps us to reach the goal
- Affective
- Very friendly and make us understand concepts easily and clearly
- The trainers teach very well . They teach practical as well as theory concepts also .
- He is very friendly and he explain good
- We can understand every topic clearly
- they come to each of us and explain in friendly way
- I don't understanding concept sir will explain the concept
- Sir support good and response well
- The mentor was helpful and made easy to understand
- They are good at explaining
- We can understand properly without any doubt

Link for the session feedback is:

feedback response of session 1,2,3

Session Glimpses: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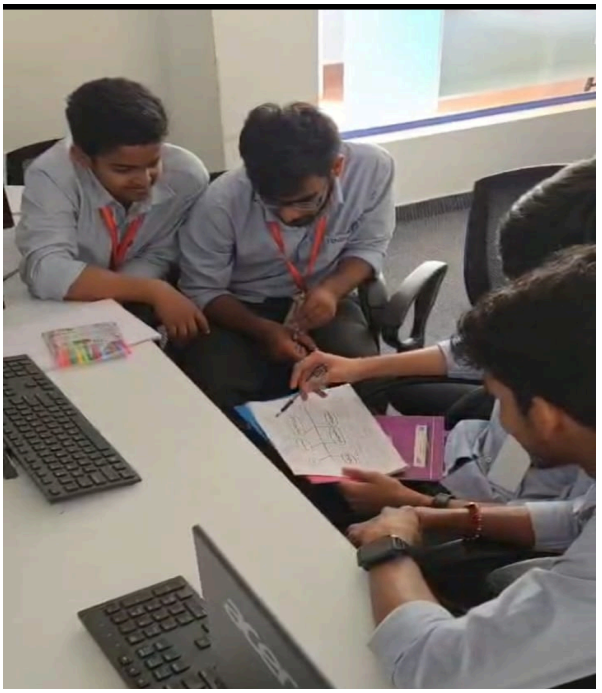


Session 4 - Introduction to Basic Electronics

Session Overview:

Session 4 focused on the fundamentals of electronics, covering key concepts such as the working principle of resistors, identification of resistor values using color codes, and the role of resistors in controlling current flow. An introduction to the breadboard was provided, including its internal connections and usage for prototyping circuits without soldering. Basic circuit diagram representation and standard symbols were also explained, along with the concept of current flow in a closed circuit.

Practical implementation included assembling and testing simple circuits on a breadboard. These experiments involved configurations such as a resistor with an LED, and a resistor with an LED and buzzer, demonstrating current limiting and component behavior. Additionally, methods for calculating and verifying appropriate resistor values for safe circuit operation were discussed and applied.

Session 4 Feedback:

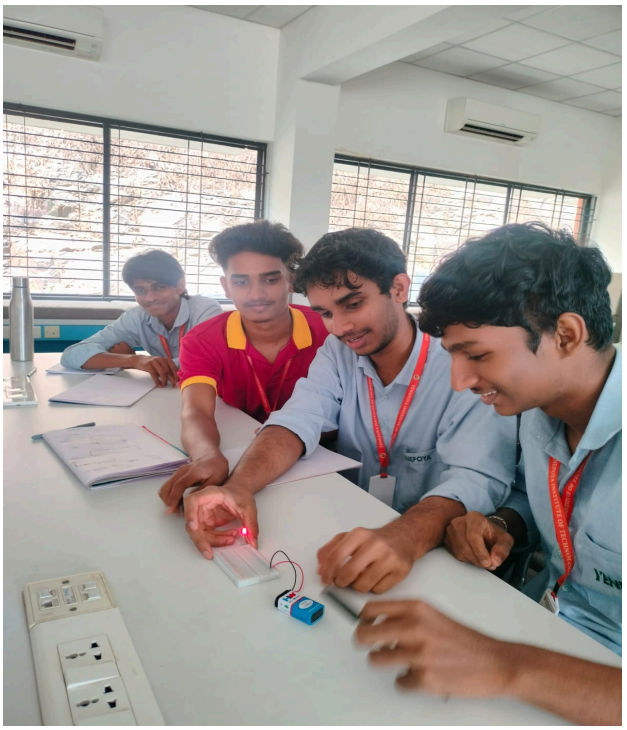
We got around 131 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

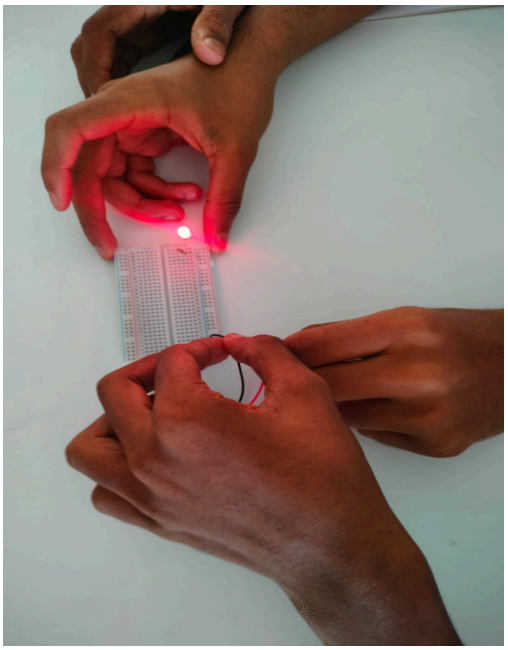
- It was an excellent class and we learn more things rather than studying only theory parts
- Good session
- Very good teaching
- Good class
- It's good interactive and I enjoyed
- We can understand properly without any doubt
- The Trainer they teach very well .
- They teach pratical as well as theory concepts also .
- They where good in explaining the activity based projects and good in interacting with all
- He maintains our torcher all the make us decipline
- He will guide us for all activities
- Explanation is good
- He makes sure everyone understood the session
- He was guide us for all activities

Link for the session feedback is:

feedback of session 4

Session Glimpses:

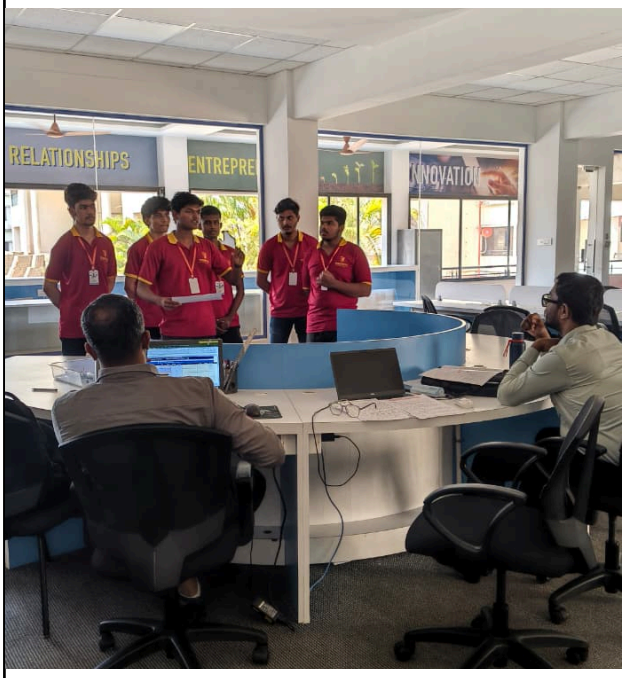




Session 5 : CIE Evaluation Phase 1 (Trial)

Session Overview: In this session, teams presented their research using mind mapping, fishbone analysis, and problem definition techniques. Instead of giving them final marks, this session was treated as a trial to highlight their mistakes, so they can improve their research for the final CIE 1 evaluation.

Session Glimpses:



Session 6 - Introduction to Microcontrollers

Session Overview:

Session 6 was focused on understanding microcontrollers like Arduino Uno and the basics of coding for embedded systems. Students learned what variables and values are, how to properly define and initialize them in code, and how to set up and configure different components using pinMode(). The session also introduced functions in Embedded C to help organize and reuse code efficiently. Additionally, students explored digitalWrite() for simple ON/OFF control of components and analogWrite() for controlling outputs like LED brightness using PWM signals, giving them a practical understanding of how hardware and software interact.

Session 6 Feedback:

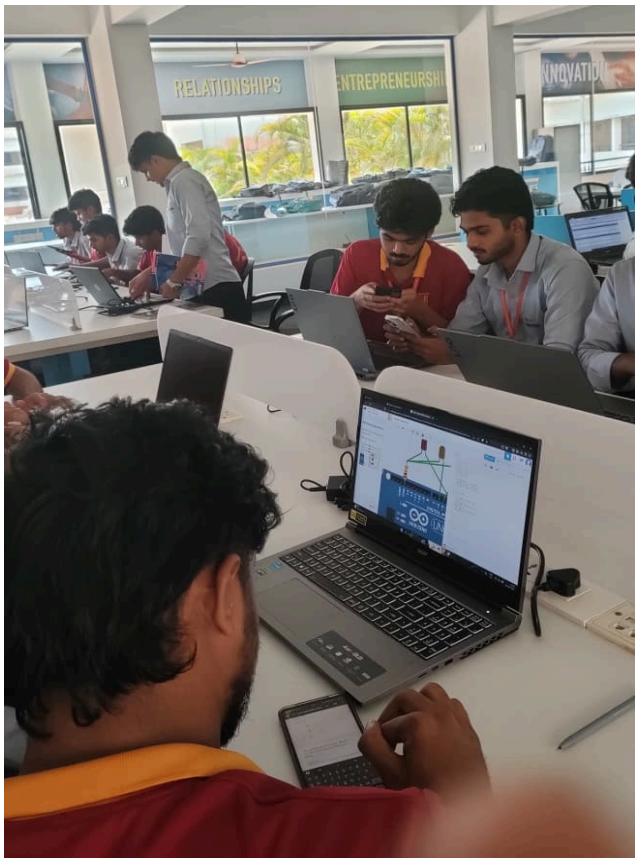
We got around 200 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

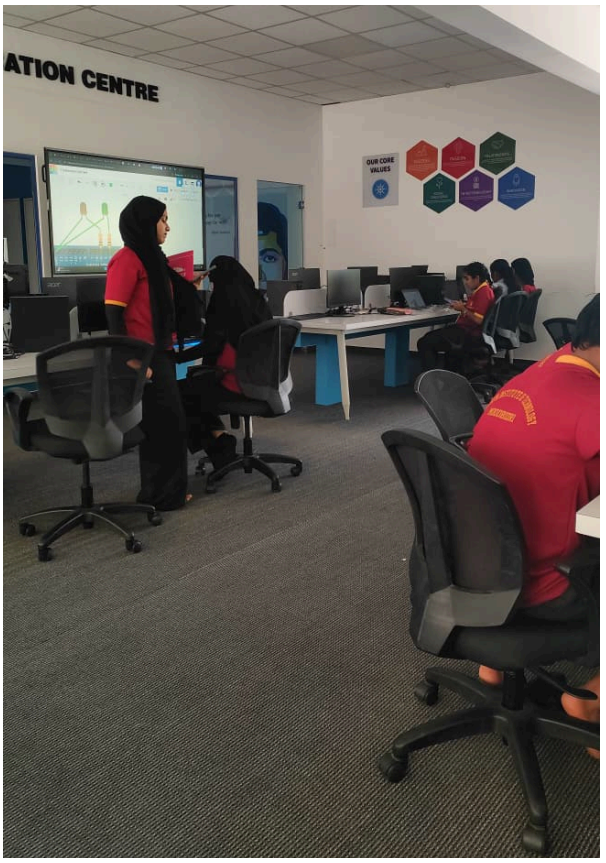
- **“Mentor explains each and everything precisely with perfection”**
- **“Perfect mentor”**
- **“Absolutely amazing We have got to learn so many things using our laptop It was great”**
- **“The mentor explained the concepts very clearly and the sessions were very helpful and interactive. I was able to understand the topics easily.”**
- **“Sir was explaining very clearly with example”**
- **“Explained and guided us properly so that we understand each and everything and do it up by our own”**
- **“All the doubts are cleared and sir explains very well”**
- **“He makes sure everyone understand the concept”**
- **“sessions were informative and the trainer explained the topics clearly. It helped me understand the concepts better”**
- **“He explains very patiently”**
- **“Well explained the concept”**
- **“Clarity about the topic”**
- **“The explanation is very good”**
- **“His explanation is good I can understand easily”**
- **“I understand very well”**
- **“I learned many new things from this class and explanation was very good”**
- **“I have learned many new concepts. Also a new process of finding solution to problems.”**
- **“Got clear picture about what to do in the upcoming future”**
- **“It's new thing for me but because of sir I learn it very easily”**
- **“Good support and information provider”**

Link for the session feedback is:

Session 6 Feedback Form

Session Glimpses:





Session 7 - Introduction to Sensor Interfacing And Analog Data Processing Using Arduino

Session Overview:

In this session, students learned how to take readings from sensors using the `analogRead()` function in Arduino. They explored how analog data is captured from sensors and how these values can be interpreted for real-world applications. Students also learned how to use `if-else` conditions to analyze and map sensor readings to meaningful outputs.

Additionally, the session covered working with the Arduino IDE, including writing, uploading, and testing code on the Arduino Uno. Students practiced simulating their circuits and code using Tinkercad, which helped them visualize the working of sensors and outputs in a virtual environment. They also learned how to display and monitor real-time sensor data using the Serial Monitor, which is useful for debugging and understanding system behavior.

Overall, the session helped students build a strong foundation in reading sensor data, applying logical conditions, and working with basic embedded systems tools and platforms.

Session 7 Feedback:

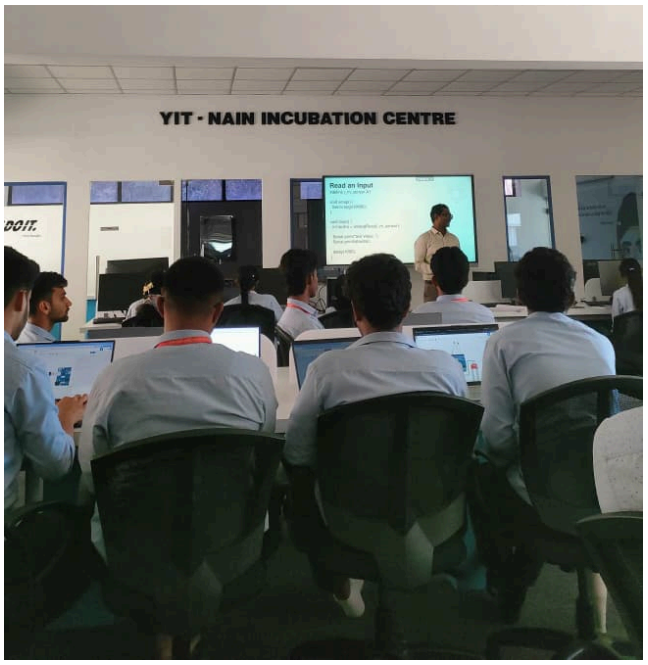
We got around 88 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

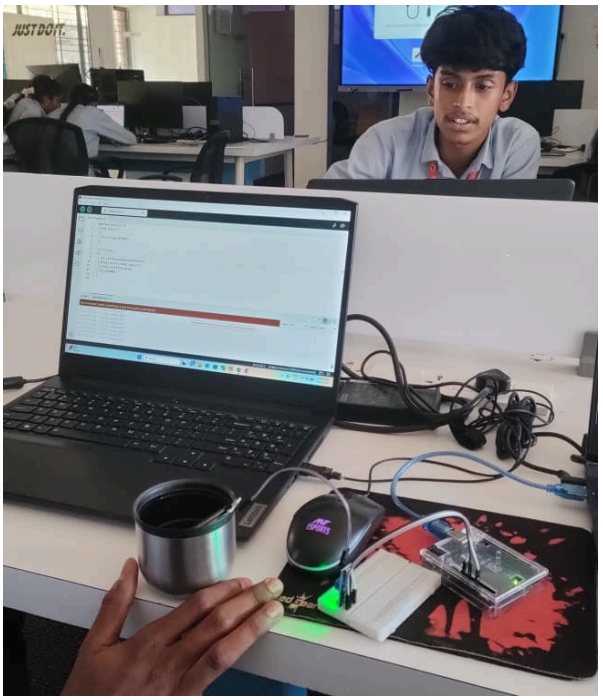
- **Well practical and theoretical explanation**
- **Nice**
- **Good mentorship**
- **New experience it was fun doing the real connections**
- **I understand that all concepts**
- **The way he explains is good**
- **Teaching is good**
- **The mentor's guidance and support made the session helpful**
- **Excellent**
- **They teach very clearly**
- **Awesome session**
- **Excellent**
- **Very good**
- **It was very good**
- **It was a very informative class**
- **The mentor explained the concepts clearly and encouraged students to participate in the discussion**
- **It was very informative class**
- **Helpful**
- **Calm, gives good explanation and makes sure students understand**
- **The explain clearly**

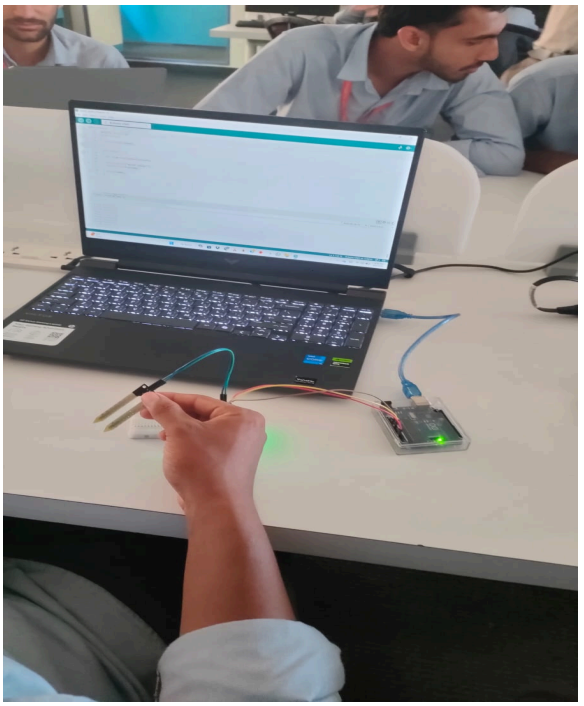
Link for the session feedback is:

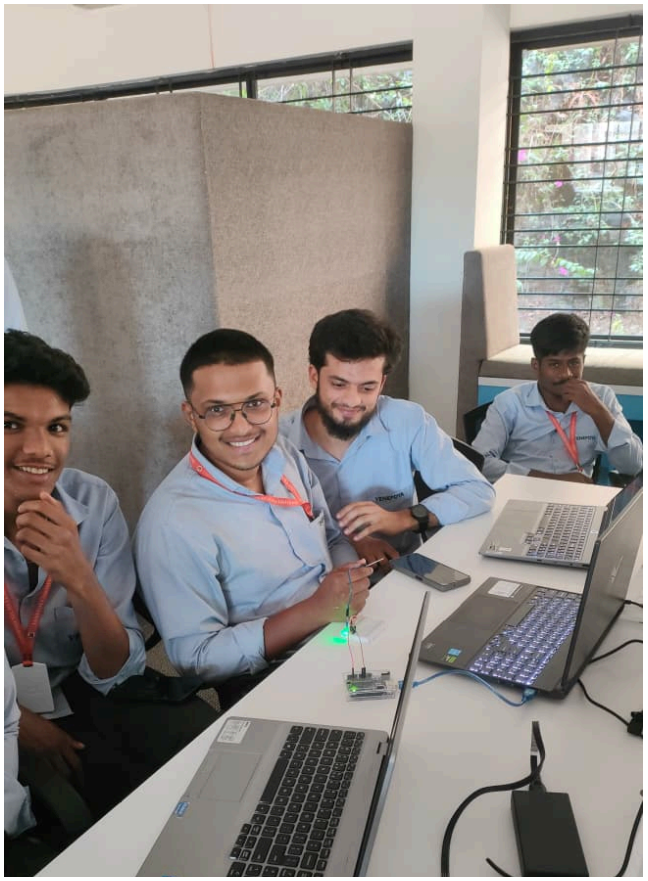
Session 7 FeedBack

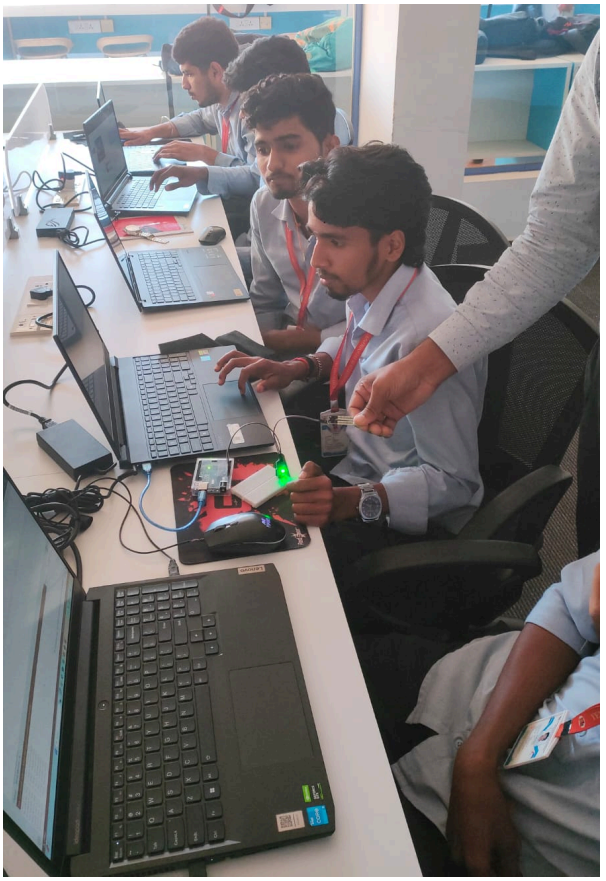
Session Glimpses:











Session 8 - Onshape CAD Designing

Session Overview:

In this session, students were introduced to CAD designing using Onshape, where they learned the fundamentals of 2D sketching including lines, circles, dimensions, constraints, and fully defining sketches. They explored important modeling features such as extrude, fillet, text sketching, linear pattern and shell. Students were also introduced to sketch relations, parametric modeling concepts, feature editing, and maintaining design accuracy using dimensions and constraints.

The session covered navigation of the software interface, creating and modifying sketches, converting 2D profiles into 3D models, and understanding the workflow from concept to model development. Students learned about design intent, symmetry in modeling, repeated feature generation using patterns, and hollow part creation using shell operations.

As a hands-on activity, students applied all these concepts to design a LEGO block, where they created the base geometry, added patterned studs, used fillets for edge finishing, engraved text using text sketching, and used shell features to understand internal geometry. Through this exercise, students gained practical exposure to product modeling, feature-based design, parametric CAD workflow, and fundamental engineering design principles.

Session 8 Feedback:

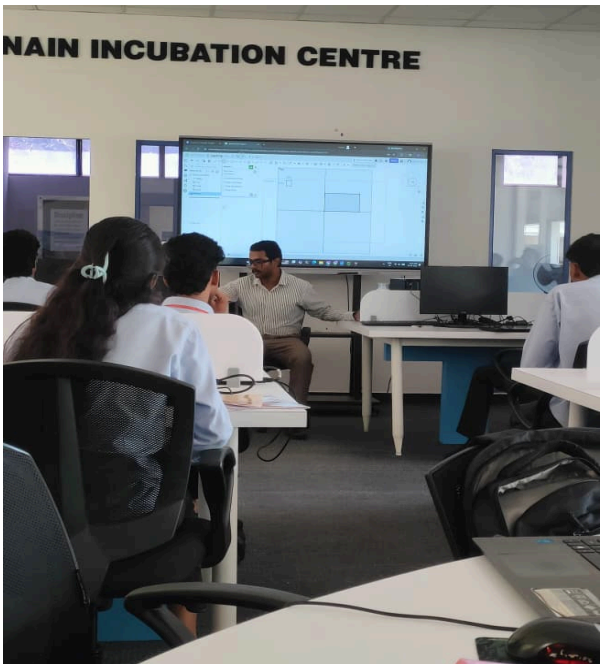
We got around 86 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

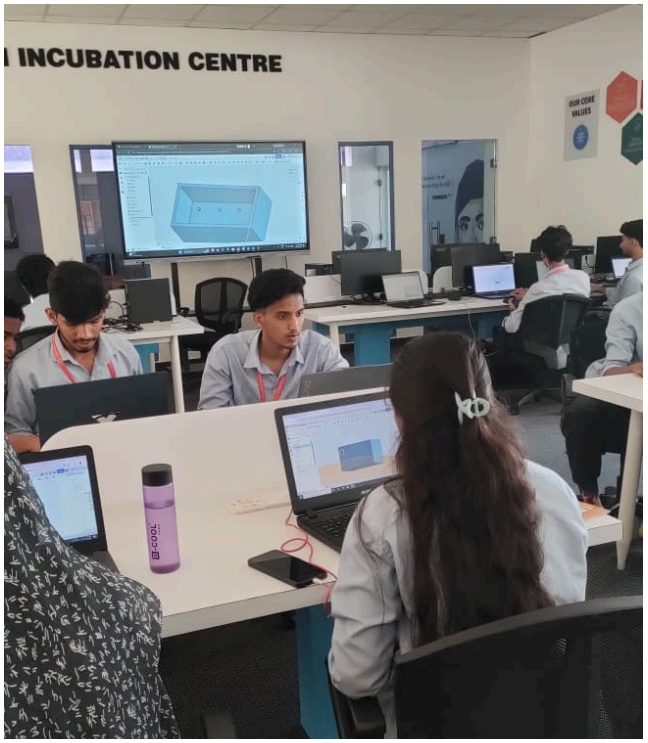
- **The class is so good that improves our knowledge and our skills**
- **Good explanation and friendly**
- **Good mentoring**
- **I understand everything from his class**
- **For his supports**
- **Teaches well**
- **Makes sure everyone understand it**
- **Very good explanation**
- **It was a creative session**
- **Explained form was very good**
- **Good explanation and learned new things**
- **Mentor is good**
- **Nice learning experience**
- **It was very informative class**
- **Helpful**
- **Calm, gives good explanation and makes sure students understand**
- **The explain clearly**

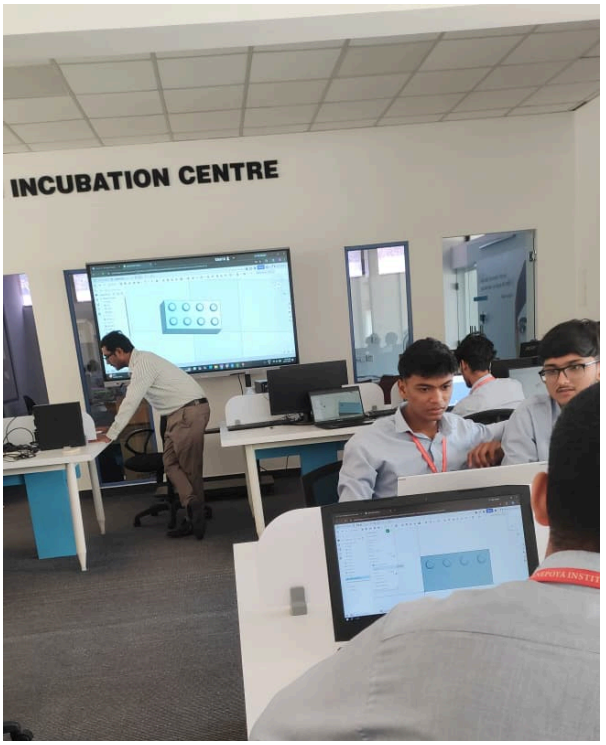
Link for the session feedback is:

Session 8(On Shape) (Responses)

Session Glimpses:







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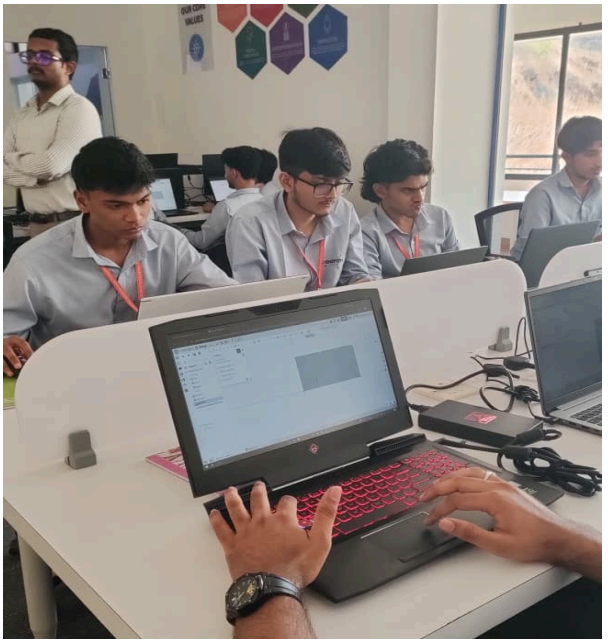


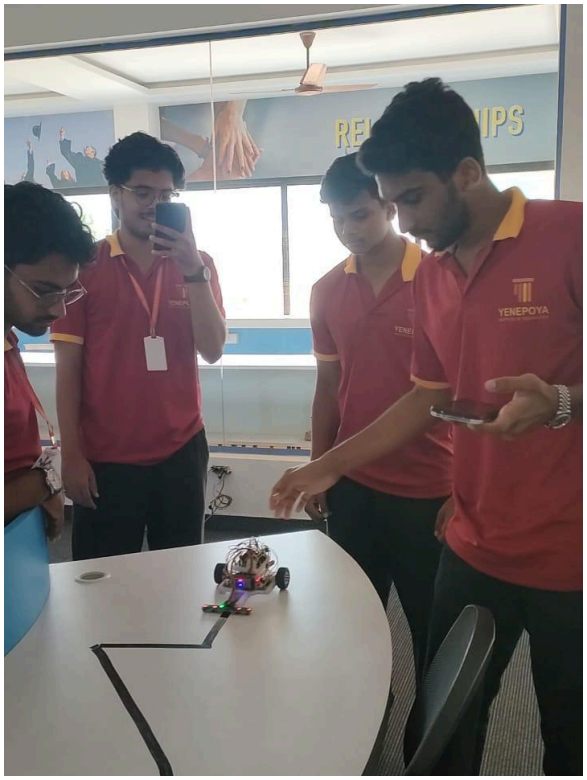
Student Participation in Technical Events (CIT):

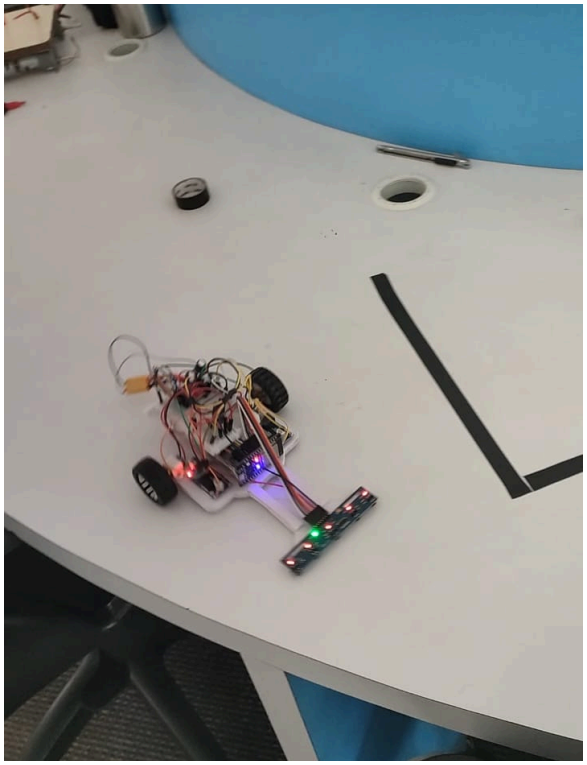
Two teams from YIT actively participated in a national-level technical event, “Robo Mania”, organized by Channabasaveshwara Institute of Technology, Tumkur. The event provided a valuable platform for students to demonstrate their practical knowledge and technical skills in robotics.

Out of the two teams:

- One team participated in the Robo Race competition.
- The other team participated in the Line Follower Robot competition.

Although the teams did not secure any prizes, their participation contributed significantly to their learning experience by enhancing their technical understanding, problem-solving abilities, and teamwork skills.





Session 9 - Ultimaker Cura - Cad Slicing & 3D Printing

Session Overview:

In this session, students were introduced to 3D printing fundamentals using Ultimaker Cura, where they learned about slicing 3D models and preparing them for printing. The session covered the purpose of slicing software, importing and positioning models, adjusting scale and orientation, and generating toolpaths for printing.

Students explored key slicing parameters such as layer height, infill density, print speed, wall thickness, supports, adhesion settings, nozzle and bed temperatures, and material selection. They also learned how these parameters affect print quality, strength, surface finish, and print time. The process of converting a 3D model into G-code for the printer was demonstrated as part of the slicing workflow.

In addition, students were introduced to the working principle of a 3D printer, including its major components such as the extruder, hotend, print bed, stepper motors, and motion systems. They learned how additive manufacturing works layer by layer, how filament is melted and deposited, and how printer movements along the X, Y, and Z axes create physical objects from digital models.

Session 9 Feedback:

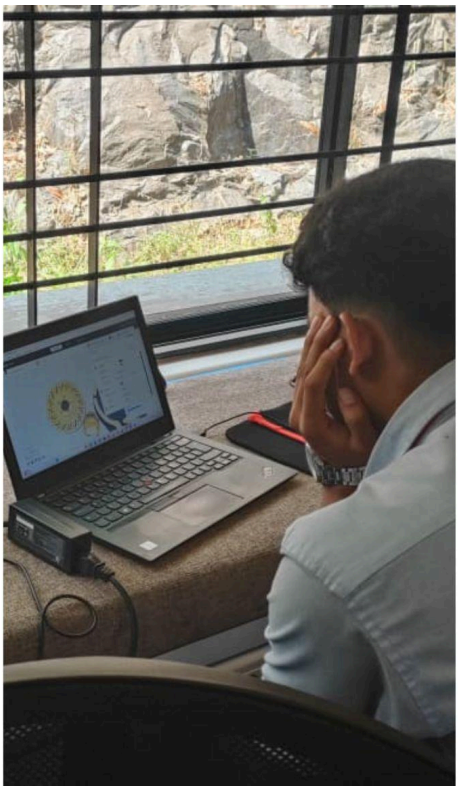
We got around 86 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

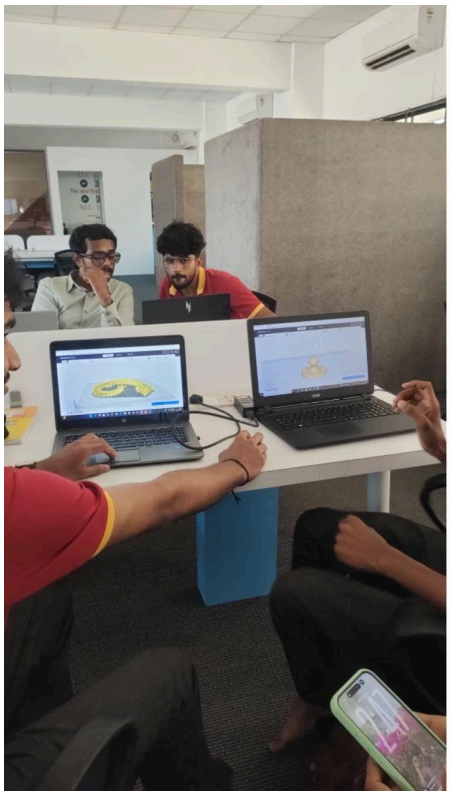
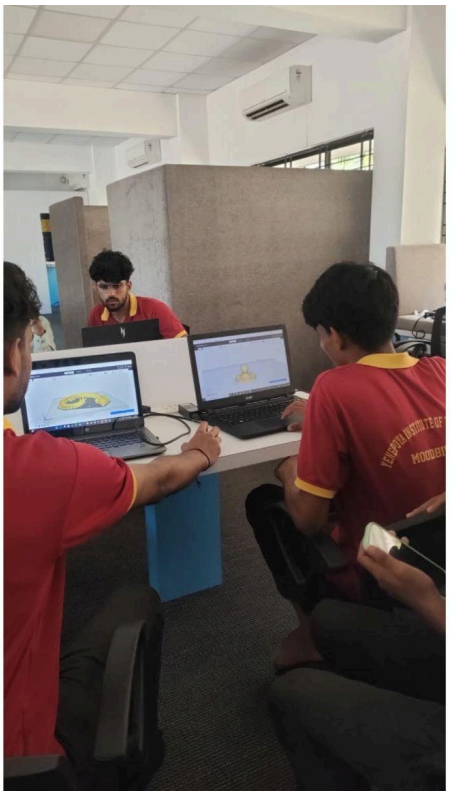
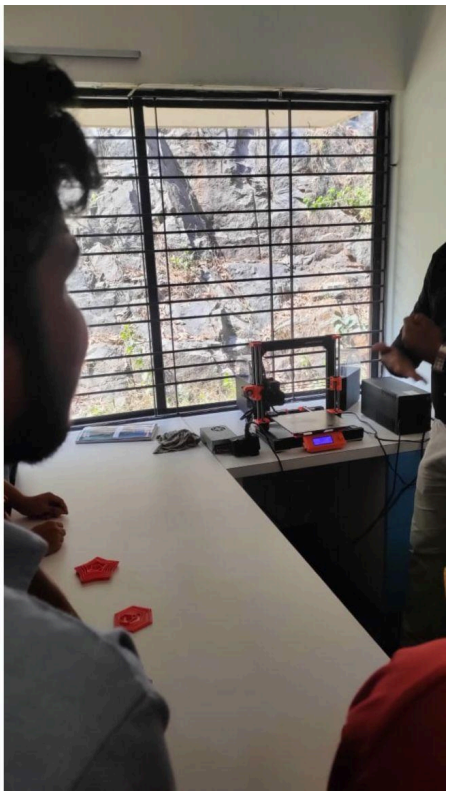
- **Because the overall experience was very good**
- **Good understanding**
- **Explain nicely**
- **The class is understandable**
- **Good teaching**
- **Teaching was good**
- **Good explanation**
- **Good mentorship**
- **Good lecturing**
- **Explaining way was very nice**
- **It was a very good explanation about the 3D printing**
- **Inspiring**
- **student friendly nice class**
- **its improving our skills**
- **Explanation good, and approachable**

Link for the session feedback is:

Session 9(3D design slicing and printing) (Responses)

Session Glimpses:



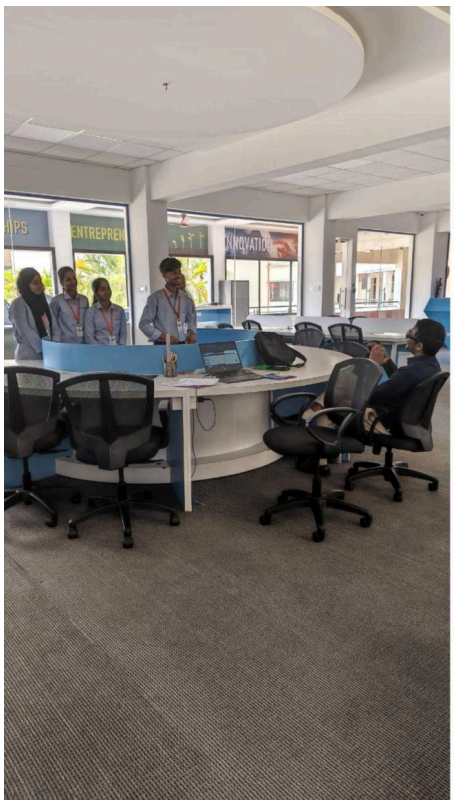




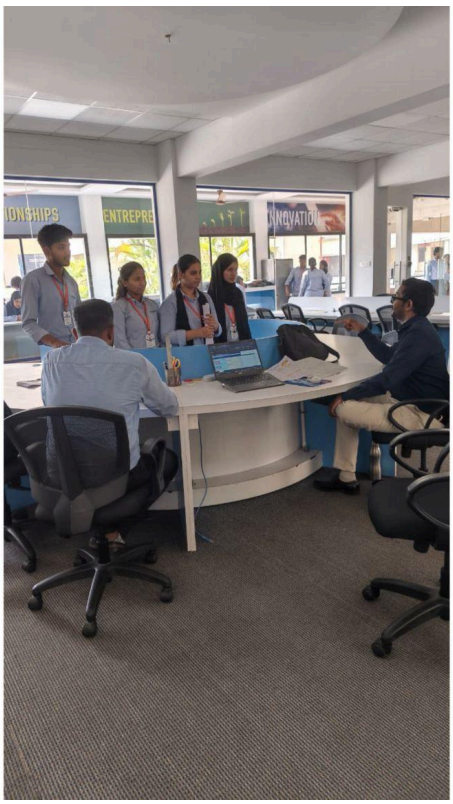
CIE Evaluation

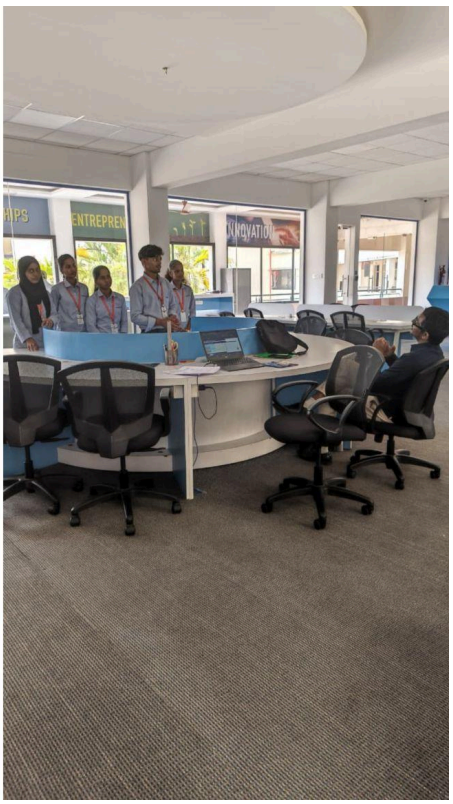
The CIE evaluation for Phase 1 commenced last week. Due to the formation of interdisciplinary groups and scheduling constraints, the evaluations are being conducted during lunch hours as well as on Fridays and Saturdays. Out of a total of 49 teams, evaluations for 20 teams have been successfully completed so far.

During the evaluation, students are presenting their Phase 1 work, explaining the concepts they have learned, the skills they have developed, and the progress they have made. They are also demonstrating their understanding through presentations, showcasing their designs, implementations, and overall learning outcomes from the phase. This process is helping assess both their technical knowledge and their ability to communicate ideas effectively.

Session Glimpses:







RELATIONSHIPS
ENTREPRE
INNOVATION

Session 10 - Antigravity - Vibe Coding

Session Overview:

In this session, students were introduced to Antigravity and learned how to use effective prompts to build production-ready software applications. The session focused on understanding prompt engineering techniques, structuring clear instructions, refining outputs, and using AI tools to accelerate software development workflows.

Students explored how prompts can be used to generate application ideas, develop functional features, improve UI/UX elements, and streamline the process of building software products. They also learned how AI-assisted development can help in rapid prototyping, debugging, and creating deployable solutions.

As a hands-on activity, students used Antigravity to create their own portfolio websites, where they applied prompt-based development techniques to design layouts, generate content, and build personalized portfolios. This session provided practical exposure to AI-driven software development and modern no-code/low-code product creation workflows.

Session 10 Feedback:

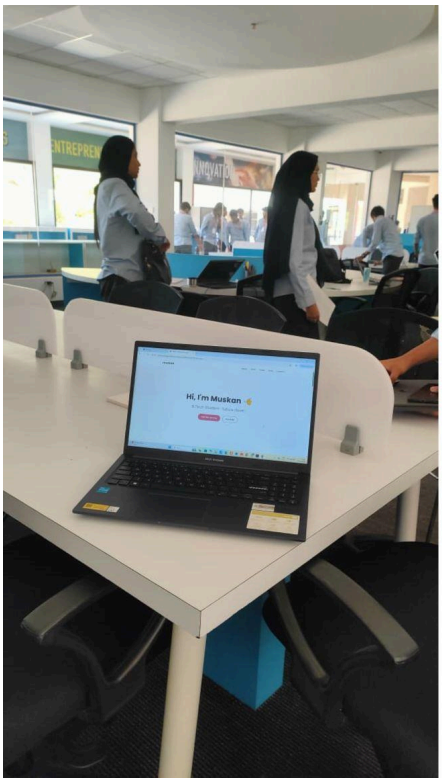
We got around 81 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

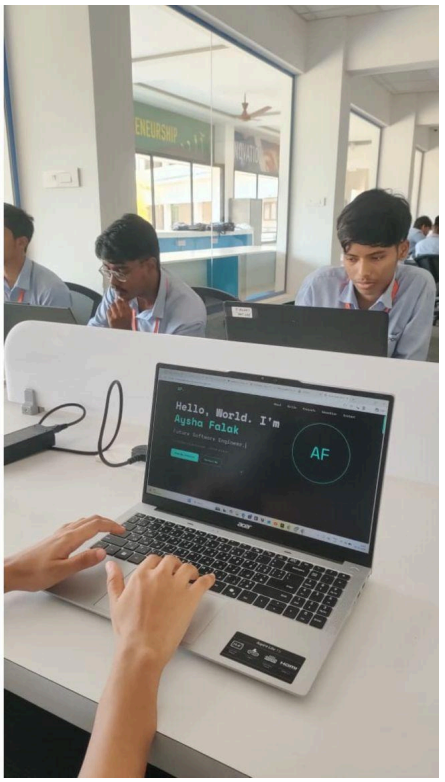
- **Good explanation.**
- **Very nice session.**
- **Good teaching and training.**
- **Nice and friendly class.**
- **Helpful and knowledgeable session.**
- **I understood the concepts well.**
- **The session was amazing.**
- **Sir was explaining properly and clearly.**
- **Good session, we learned new things related to Claude and gained useful knowledge.**
- **Explains well and makes the class easy to understand.**
- **It was an excellent session.**
- **Cool and informative session.**
- **Good class with simple and clear explanations.**
- **The session was very useful and helpful.**
- **Nice explanation and good interaction.**
- **Super session, learned something new.**
- **Good teaching style and clear communication.**
- **It was a very good and understandable session.**

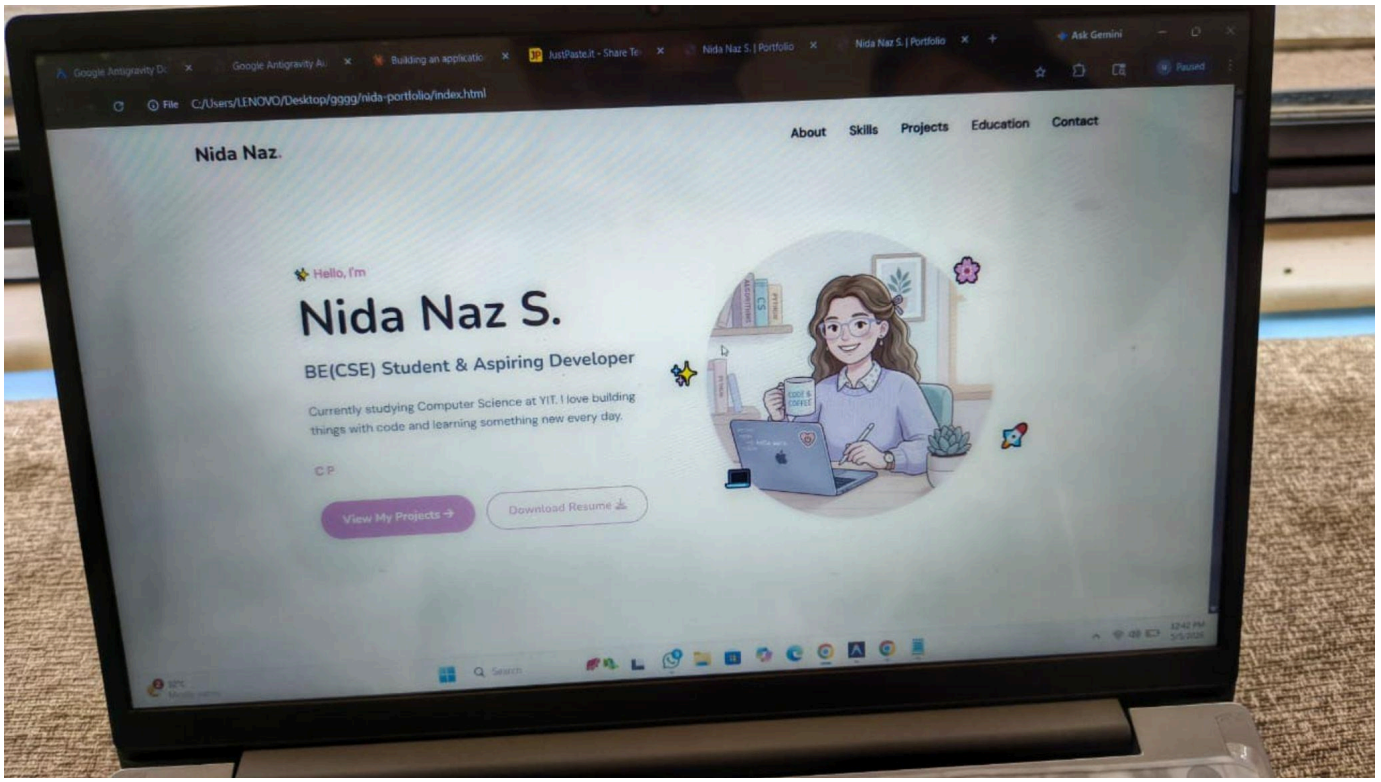
Link for the session feedback is:

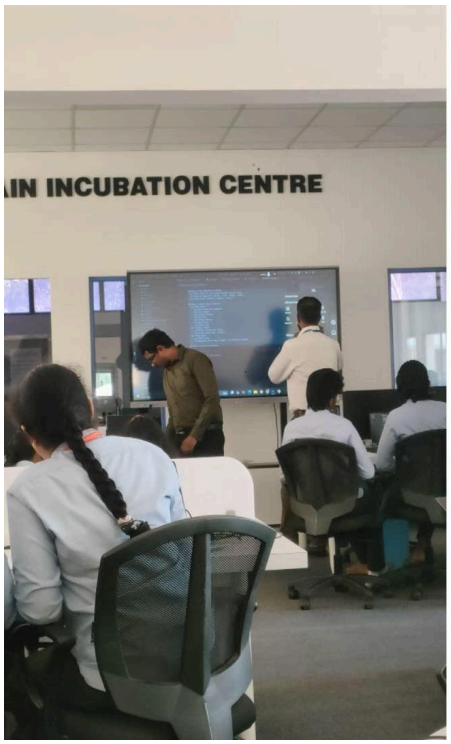
Copy of Copy of Session 10(vibe coding) (Responses)

Session Glimpses:





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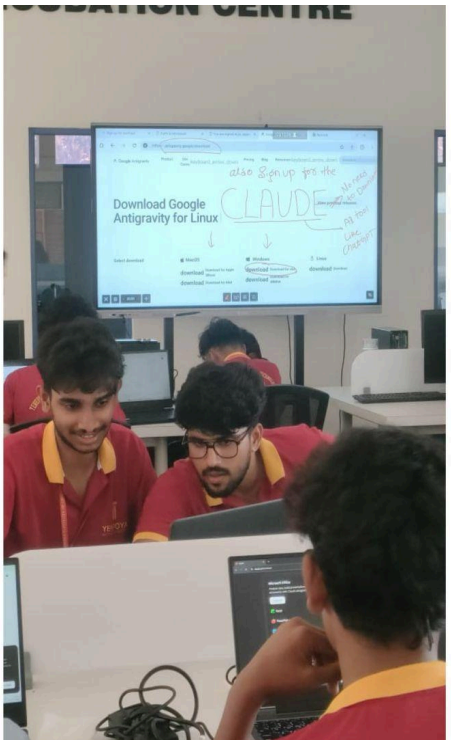
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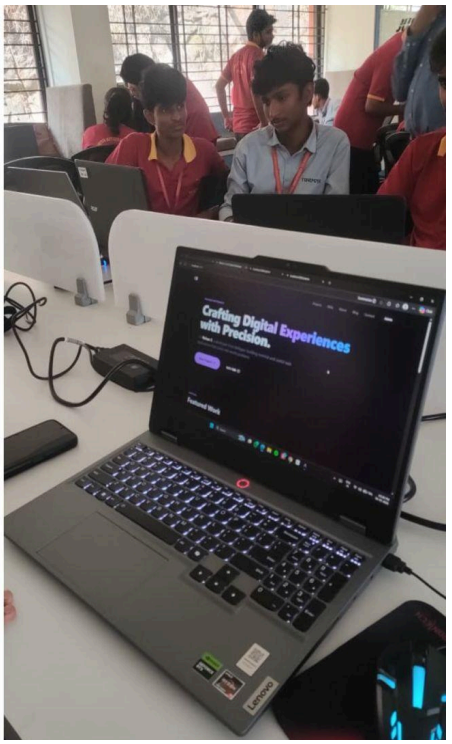
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Session 11 - EasyEDA: PCB Designing

Session Overview:

In this session, students were introduced to PCB designing using EasyEDA. They learned the fundamentals of electronic circuit design, including how to create schematic diagrams by selecting and connecting electronic components properly.

The session covered the use of net ports and wiring techniques for making organized electrical connections within the schematic. Students also learned about component placement, labeling, and verifying circuit connections before proceeding to PCB design.

After completing the schematic, students were guided through the process of converting the circuit into a PCB layout. They explored PCB design concepts such as board dimensions, component arrangement, routing tracks and layer management for efficient circuit design.

Finally, students learned how to generate Gerber files from the PCB layout, which are used for PCB fabrication and manufacturing. The session provided students with a complete understanding of the workflow from schematic creation to PCB layout design and preparation for real-world PCB production.

Session 11 Feedback:

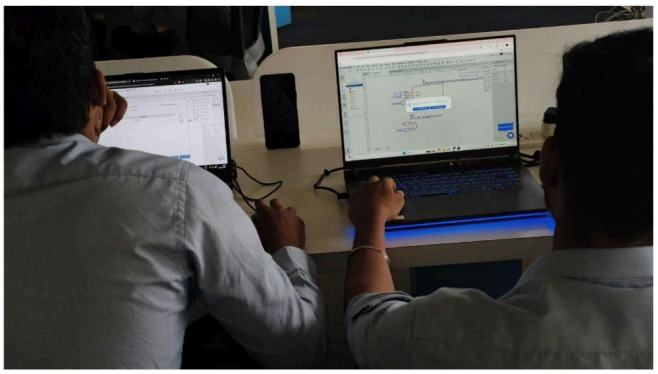
We got around 115 responses from the students regarding feedback. Some of the feedbacks from the students itself have been mentioned below

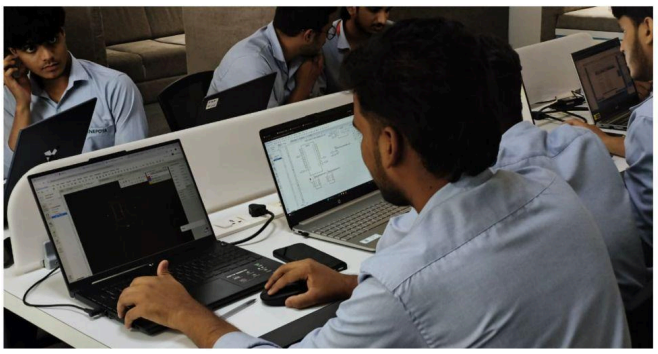
- **“Good teaching”**
- **“Amazing”**
- **“Explain very nicely”**
- **“Very good session”**
- **“Session was fun and understandable .”**
- **“The content was very practical and easy to follow.”**
- **“Great insights and real-world examples.”**
- **“Explanation was good”**
- **“Intresting class”**
- **“It was informative and helpfully for our project”**
- **“Got learn about pcbs useful for future projects”**
- **“I understood everything clearly”**
- **“He teaches very good”**
- **“They helped a lot..solved our doubts and gave a great experience”**
- **“They are good at explaining”**
- **“The explanation is good”**
- **“Useful and Informative.”**
- **“The explanation was interesting and helping us with our learning”**
- **“Helpful and good explanation”**
- **“Superb excellent teaching”**

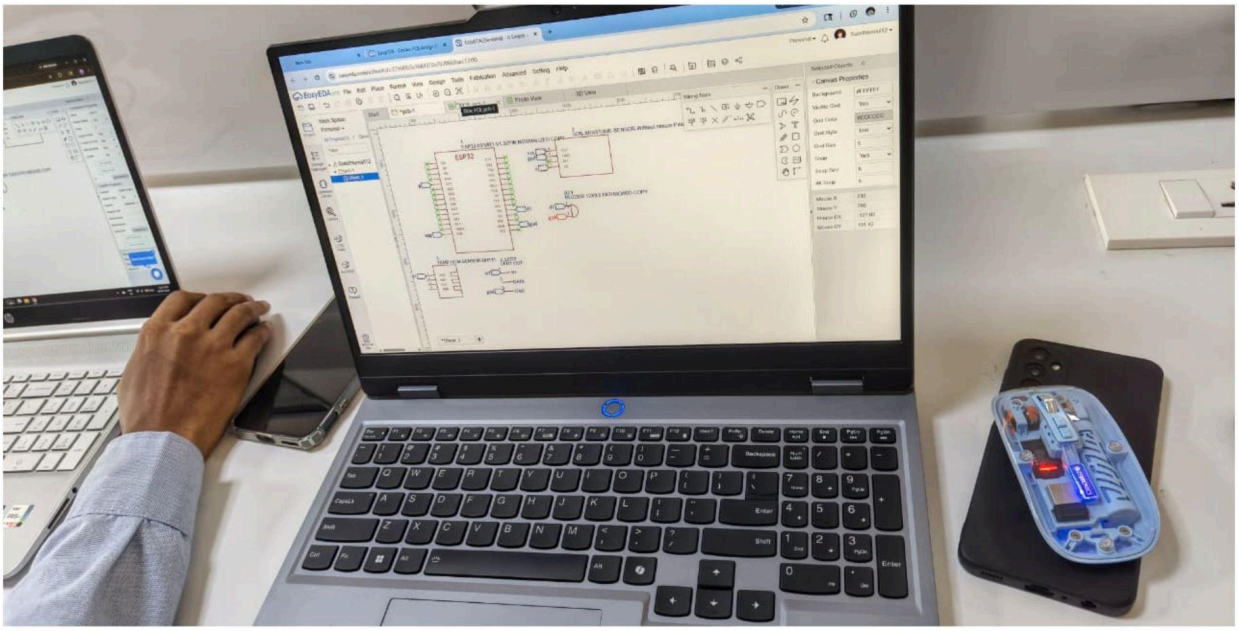
Link for the session feedback is:

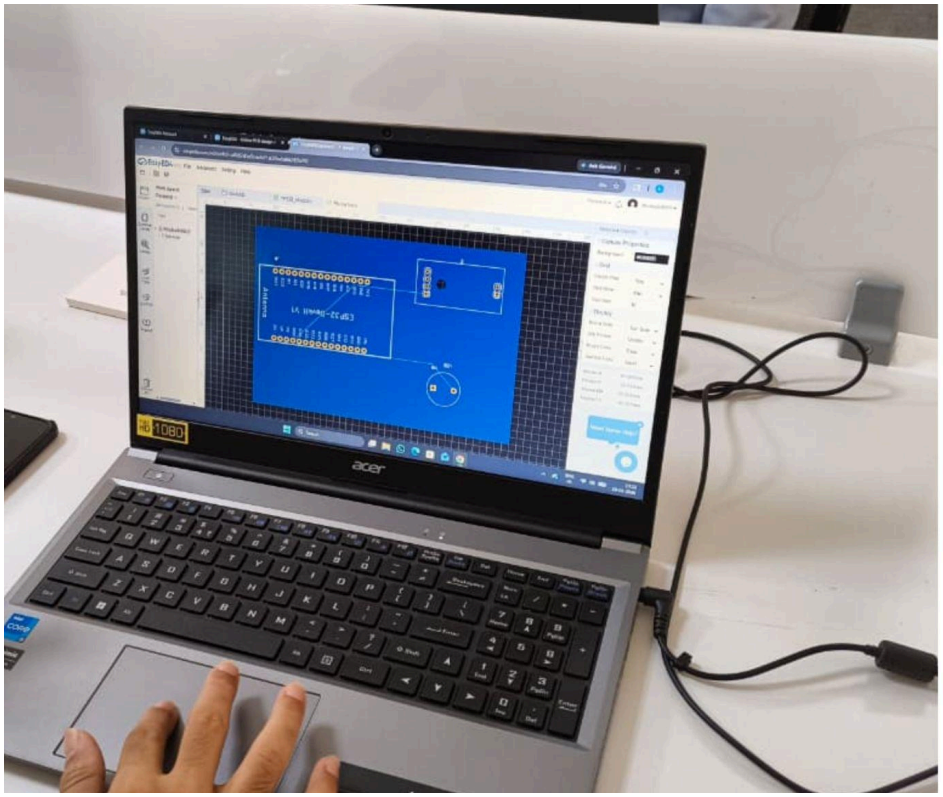
Bonus Session PCB Designing (Responses)

Session Glimpses:





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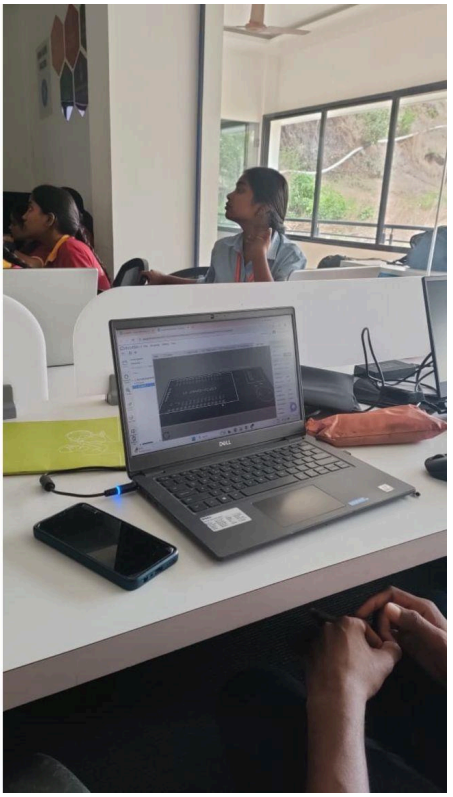
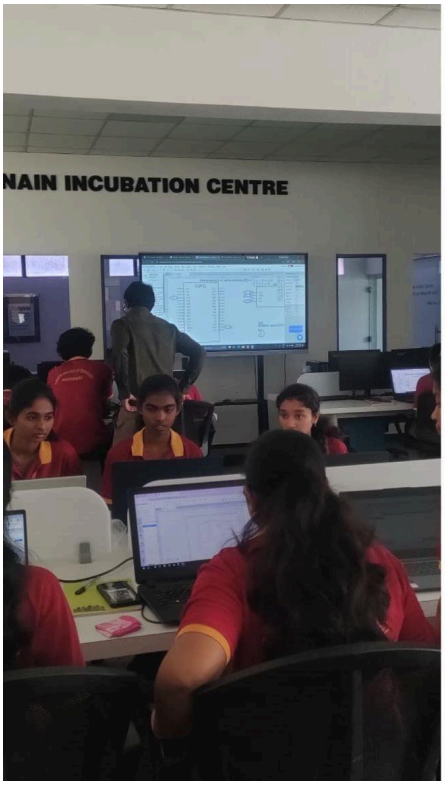
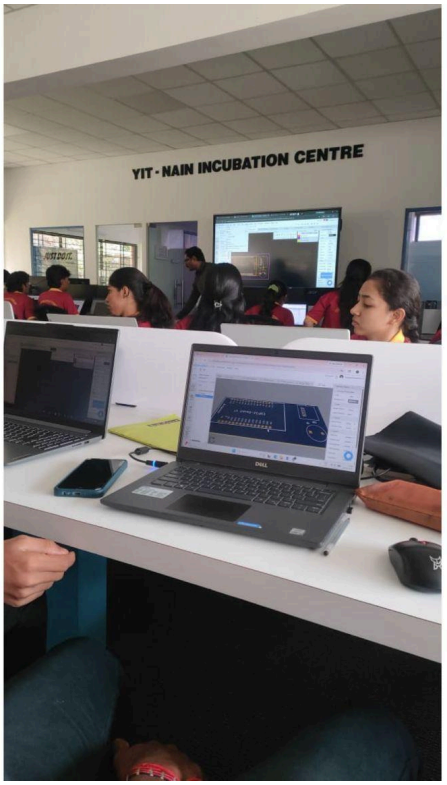
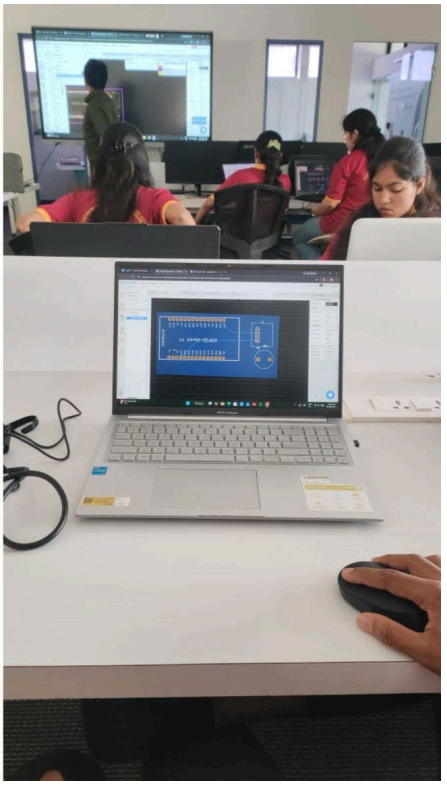



NAIN INCUBATION CENTRE


YIT - NAIN INCUBATION CENTRE



Student Participation in Technical Events (PES Mandya & VVCE):

Two teams from YIT actively participated in a technical event, “Line Follower”, “Robo Race” & “Robo Soccer” organized by PES Mandya. The event provided a valuable platform for students to demonstrate their practical knowledge and technical skills in robotics.

One team from YIT actively participated in a technical event, “FusionX 1.0 Hackathon” organized by VVCE. The event provided a valuable platform for students to demonstrate their practical knowledge in solving problems related to the real world.

Out of all these competitions, one team secured the 3rd Runner-Up position at PES Mandya. Although the other teams did not win any prizes, their participation contributed significantly to their learning experience by enhancing their technical understanding, problem-solving abilities, and teamwork skills.

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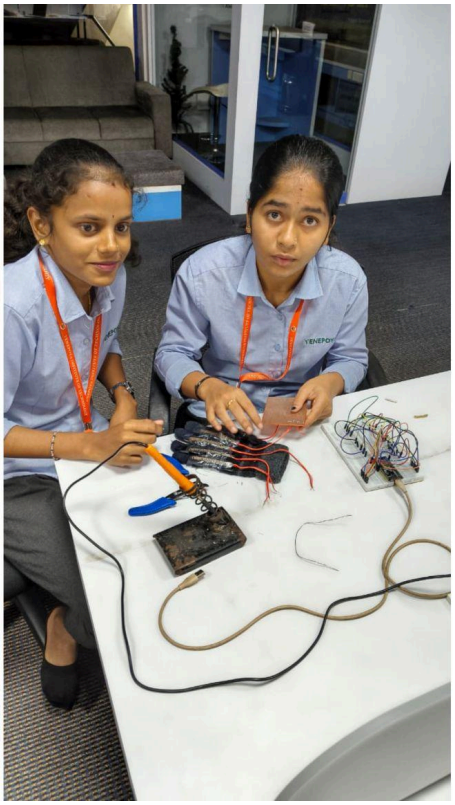
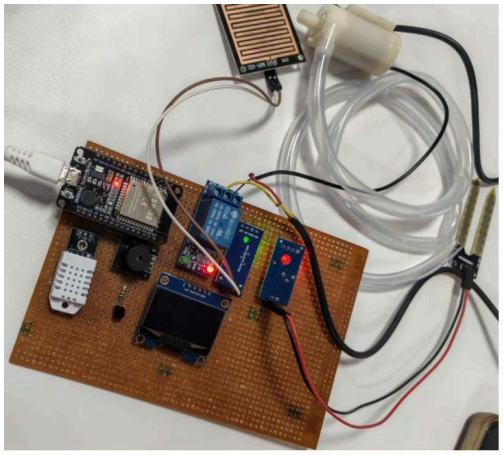
Prototyping Day 1 & 2

During Prototyping Day 1 and Day 2, most of the student teams actively participated by seeking guidance and mentorship for their hardware and software projects in preparation for the Demo Day presentations. We provided technical support, reviewed project progress, helped students troubleshoot challenges, and offered suggestions for improving both the functionality and presentation of their projects.

These sessions enabled students to refine their prototypes, validate their ideas, and enhance their confidence before the final demonstration. The collaborative environment encouraged problem-solving, teamwork, and effective project planning, ensuring that teams were better prepared to showcase their work on Demo Day.





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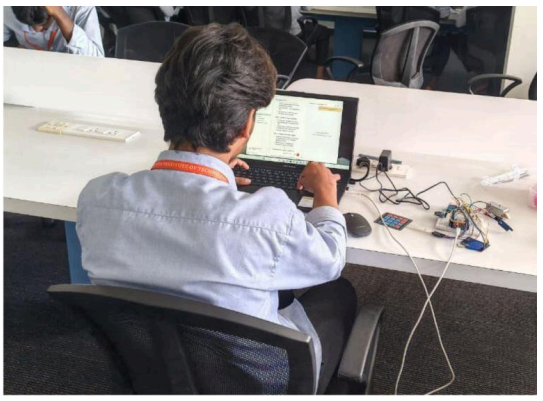

Demo Day

During Demo Day, a total of 288 students, organized into 49 teams, presented their projects showcasing the knowledge and skills they had developed throughout the program. The presentations included a diverse range of hardware projects, software applications, and integrated hardware–software solutions designed to address real-world problems and innovative use cases.

Each team demonstrated the functionality of their prototypes, explained the design and development process, highlighted the technologies and tools used, and discussed the challenges encountered during implementation. The event provided students with an opportunity to exhibit their technical expertise, creativity, teamwork, and problem-solving abilities while receiving valuable feedback from mentors, faculty members, and evaluators. Demo Day served as a platform to celebrate students' achievements and the successful culmination of their project development efforts.

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Link for the Demo Day Photos:

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